

Full Length Article

Mining waste-driven carbon capture via ocean alkalinity enhancement[☆]

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ABSTRACT

Ocean alkalinity enhancement (OAE) has emerged as a promising strategy to mitigate ocean acidification and reduce global warming. Traditional metal (e.g., critical materials) mining industries release alkaline waste via mining tailings with high concentrations (99.2%) of calcium and magnesium oxide (CaO, MgO). Incorporating mining waste into OAE processes is less energy intensive than processes relying on calcination of limestone for CaO production. The solubility limit of simulated mining waste in American Society for Testing and Materials (ASTM) seawater is 75 mg·L⁻¹, which can sequester 118 mg of carbon dioxide (CO₂). The solubility in seawater retrieved from Sunset Beach, FL was 25 mg·L⁻¹. Changes in pH, total alkalinity, and total inorganic carbon were analyzed to confirm the successful addition of simulated alkaline mining waste without the formation of secondary precipitation. This study proposes a new OAE strategy where a facility is developed nearby ocean waters that mixes alkaline waste with seawater. Subsequently, the seawater is met with previously captured, pure CO₂ to bring the pH back to 8.2 and eliminate the risks of pH shock and secondary precipitation. Technoeconomic analysis estimated an energy requirement of 1.4 GJ per ton of CO₂ stored that resulted in a processing cost of \$266 per ton of CO₂ sequestered (4.2 GJ per ton, \$807 per ton of CO₂ for real seawater). Results from this study underscore the potential for utilizing mining waste in OAE processes and provide a pathway for practical deployment.

Introduction

While climate change continues to evolve, the ocean acts as a natural sink for carbon dioxide (CO₂) and has absorbed approximately 30% (139,000 Gt) of all CO₂ emissions since the Industrial Revolution. (Renforth and Henderson, 2017) On an annual basis, the ocean absorbs 12.5 Gt of the 37.4 Gt of CO₂ emitted worldwide. Absorption of CO₂ in seawater leads to the formation of carbonic acid (H₂CO₃), a weak acid that dissociates, leading to increased concentrations of hydrogen ions that have decreased the ocean's pH from 8.2 to 8.1. (Renforth et al., 2013) This seemingly subtle 0.1 decrease corresponds with a ~30% increase in acidity that is detrimental to marine life. (Foteinis et al.,

2022; Burger et al., 2022) As a result of this “ocean acidification,” available carbonate ions bond with excess hydrogen ions, leaving less carbonate ions available for calcifying organisms (e.g., oysters, corals) to build and maintain their shells, skeletons, and other calcium carbonate (CaCO₃) derived structures. (Hu et al., 2024) Further reduction in seawater pH could lead to shell dissolution in some organisms. (Bednarsek et al., 2022) Although the rate at which organisms are being affected has been disputed across recent studies, ocean acidification remains a detriment to marine life and needs to be addressed. (Duarte et al., 2020; Sundin, 2023; Cohen-Rengifo et al., 2023) If atmospheric CO₂ concentrations continue to increase at the current rate, seawater pH could reduce to 7.8 by the end of the current century, potentially leading

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to the extinction of many species. (Sundin, 2023)

The ocean acidification phenomenon has created research efforts in the direct ocean capture (DOC) space aimed towards removing CO₂ from seawater. (Karunaratne et al., 2025) The concentration of oceanic CO₂ is 2.38 mM, largely in the form of bicarbonate ions (HCO₃⁻, ~90%), carbonate ions (CO₃²⁻, 10%), and a small percentage of aqueous CO₂ (< 1%). Interestingly, the concentration of CO₂ in the atmosphere is 0.017 mM, a factor of ~140 times less than seawater concentrations. (Yan et al., 2022; de Lannoy et al., 2018) This stark difference in concentration has driven research towards decreasing marine CO₂ concentrations to restore its original pH from 200 years ago and protect aquatic life. Overall, this would also allow the ocean to continue performing nature's work of absorbing anthropogenic CO₂ while combating global warming.

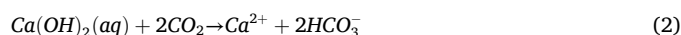
Electrochemical methods are the most mature technology in the DOC space and have been extensively studied. (Yan et al., 2022; Digdaya et al., 2020; Kim et al., 2023; Eisaman et al., 2012) Although many promising investigations have been reported, these technologies are held back by factors such as membrane fouling, electrode poisoning, scaling issues, and generation of unwanted byproducts in the form of hydrochloric acid and chlorine gas that can pose health and environmental issues. (Yafiee et al., 2024; Nawaz et al., 2023) They also have large energy requirements in the range of 3.6 to 9.0 GJ per ton of captured CO₂. (National Academies of Sciences, Engineering, and Medicine, 2021)

Alternative approaches have designed membrane-solvent absorption systems to directly remove aqueous CO₂ from seawater. (Rivero et al., 2023; Cho et al., 2025) In one study, hollow fiber membranes and sodium hydroxide (NaOH) were used to remove aqueous CO₂ with a processing cost of \$9565 per ton of captured CO₂. (Rivero et al., 2023) Another investigation utilized hollow fiber membranes with various amine-based solvents. (Cho et al., 2025) While both studies provided a different approach to membrane-based DOC, their processing costs and assumed-to-be high energy requirements are not practical for very little aqueous CO₂ (< 1%) removal. These technologies cannot remove HCO₃⁻ and CO₃²⁻ ions that represent ~99% of dissolved inorganic carbon (DIC) in seawater unless acid is introduced to lower system pH. Amine-based sorbents are also widely studied and knowingly require high temperature and energy during the regeneration process. (Wu, Zhang and Wu, 2024)

A different approach to address ocean acidification that has recently emerged is ocean alkalinity enhancement (OAE). The fundamental objective of OAE is to leverage the ocean's vast surface area to sequester atmospheric CO₂ for permanent marine storage on the order of 1000s of years. (Fakhraee et al., 2023; Sherman, 2024) By adding benign alkaline compounds to ocean waters, the pH is directly raised, enhancing the capacity for further CO₂ sequestration. Typical compounds utilized in OAE research are calcium oxide (CaO), (Renforth et al., 2013; Kowalczyk et al., 2024; Varliero et al., 2024; Moras et al., 2022) sodium hydroxide (NaOH), (Ringham et al., 2024) and magnesium oxide (MgO). (Yang, Leonard and Langdom, 2023; Moras et al., 2024) One of the biggest hurdles of this emerging field is identifying a sustainable source of alkaline materials in large enough quantities to sequester marine CO₂ on a gigaton scale. (National Academies of Sciences, Engineering, and Medicine, 2021; Sherman, 2024) Enhanced rock weathering is another potential method for increasing marine CO₂ storage that has been investigated. (Xu and Reinhard, 2025; Li, Planavsky and Reinhard, 2024) On an annual basis, silicate rock weathering removes ~0.26 Gt of atmospheric CO₂. (Hartmann et al., 2009) Interestingly, marine CO₂ storage is the least reported strategy to global warming in the literature. (Renforth and Henderson, 2017; Bellamy et al., 2012)

Most OAE literature to date utilizes CaO for marine CO₂ storage. (Kowalczyk et al., 2024; Varliero et al., 2024; Moras et al., 2022) One of the biggest advantages of using CaO for OAE is the two-to-one molar ratio of sequestered CO₂ to calcium hydroxide (Ca(OH)₂), (Renforth and Henderson, 2017) as shown in Eqs. (1) and 2. Although stoichiometry indicates a 2:1 ratio, simulations have predicted the ratio being in the

range of 1.6 to 1.8, depending on temperature. (Renforth et al., 2013) CaO is predominantly generated through calcination of limestone into lime. (Sherman, 2024; Kowalczyk et al., 2024) Limestone calcination is the conventional method of generating lime in the concrete industry, with a yearly production of 16 Mt in 2023 in the U.S. (420 Mt globally) valued at \$2.6 billion. (Apodaca, 2025) However, calcination requires a high amount of energy (~1000 °C) (Kowalczyk et al., 2024) and emits additional CO₂ during the process that is counterintuitive to downstream carbon capture. (Renforth and Henderson, 2017; Renforth et al., 2013) This additional CO₂ is emitted on an equimolar basis and reduces the potential CO₂ capture by half, as shown in Eq. (3). Identifying a source of CaO that does not emit CO₂ and takes full advantage of sequestering two moles of CO₂ would accelerate OAE research and its potential for anthropogenic carbon capture.



Some OAE investigations have considered MgO as an alternative to CaO since magnesium carbonate (MgCO₃) calcines at a much lower temperature (600 °C) and has a smaller heat of dissociation. (Yang, Leonard and Langdom, 2023; Moras et al., 2024; Renforth and Kruger, 2013) While this may reduce the energy cost associated with calcination, the process still emits additional CO₂. Additionally, only 11 Gt of MgCO₃ reserves exist on earth, and the current industry only calcines 11 Mt annually. (Renforth and Kruger, 2013) Another material being investigated for OAE is slag from iron and steel making processes. (Renforth, 2019) One study investigated steel slag for CO₂ capture from flue gas and determined an energy requirement of 1.02 GJ per ton of CO₂ while not taking secondary precipitation into consideration. (Li et al., 2020) Slag will continue to create research interest in the OAE space, but is currently held back by the wide range of compositions and risk of trace elements leaching into the environment that negatively impact marine life. (Moras et al., 2024; Guo et al., 2024)

Interestingly, traditional metal (e.g., critical materials) mining industries working to recover rare earths (neodymium, praseodymium) are producing waste via mining tailings that contain several alkaline materials including CaO and MgO. These tailings are a significant environmental concern for local communities and could serve as a major source of alkaline materials that are otherwise considered waste. Utilizing alkaline components from these tailings to perform OAE is potentially less energy-intensive than traditional calcination-based ocean liming.

Once the CaO feedstock is acquired, the most common strategy is driving a shipping freighter across the ocean and slowly releasing the material. (Renforth and Henderson, 2017; Renforth et al., 2013; Kowalczyk et al., 2024) There are many factors, such as currents and weather, that create a lack of control over the complete timeline of this process. In order to effectively carry out CaO ocean liming, the particles must fully dissolve within the first 100 to 150 m of the ocean's surface. (Renforth et al., 2013; Renforth and Kruger, 2013) Dissolution must occur at these depths to impact CO₂ levels at the surface of the ocean, and going any deeper will have little to no effect. Particle size will affect dissolution rates, requiring additional energy to achieve sufficiently smaller diameters. Discharge rate also plays a major role in the effectiveness of this strategy when considering risks such as pH shock and secondary precipitation. (Caserini et al., 2021) Adding too much alkalinity to seawater can lead to secondary precipitation, which is one of the biggest challenges of this field. (Moras et al., 2022; Ringham et al., 2024) Precipitation leads to the formation of carbonate crystals (aragonite, (Caspi et al., 2005) halite, brucite (Shaw et al., 2025)) that leads to a decrease in pH and lack of CO₂ absorption capacity. (Zeebe and Wolf-Gladrow, 2001) It has been extensively studied, with recommended concentrations for CaO additions ranging from 20 to 25 mg·L⁻¹.

(Kowalczyk et al., 2024; Varliero et al., 2024; Moras et al., 2022)

This paper investigates the dissolution of simulated mining waste in ASTM seawater and real ocean water retrieved from Sunset Beach, FL. The simulated mining waste has the composition of alkaline waste released from the Pea Ridge Mine in Washington County, Missouri. Upon full dissolution, changes in total alkalinity (TA) and total inorganic carbon (TIC) were evaluated. These additions were carried out at ambient temperature, as well as colder temperatures (5, 10 °C) to leverage CaO's retrograde solubility and increase the CO₂ absorption capacity of seawater. Absorption of CO₂ in seawater was carried out at ambient conditions (20 °C) as seen in other studies. (Varliero et al., 2024) The alkaline enhanced solutions were also injected with pure CO₂ to expedite the absorption process.

A technoeconomic analysis (TEA) was developed for a proposed mixing facility that processes 50,000 short tons of mining waste, capable of storing 71,200 t of CO₂ annually. Installing a mixing facility along coastal waters allows for tailored control over the dissolution of mining waste with seawater to avoid pH shock and secondary precipitation. After complete dissolution, the alkaline enhanced seawater is met with pure CO₂ to bring the pH back to 8.2 before being released back into the ocean. This closed loop system of mixing and absorption is more cognizant of rapid changes in acidity on aquatic life and eliminates the uncertainty that comes with releasing hydroxides into open ocean waters. This analysis projected an energy requirement of 1.4 GJ per ton of stored CO₂, along with a processing cost of ~\$266 per ton of CO₂. When this same process was performed in real seawater from Sunset Beach, FL, the energy requirement rose to 4.2 GJ per ton of stored CO₂, along with a processing cost of ~\$807 per ton of CO₂. Our estimations predict that the Pea Ridge Iron Mine (PRM) alone can provide 2.6 million tons of alkaline material, which could remove about ~4.08 million tons of atmospheric CO₂. Injecting pure CO₂ from another capture process (direct air capture, point source) provides an alternative to compressing unusable gas into an underground class VI well. Results from this study underscore the importance of identifying sustainable sources of alkaline materials to build OAE into a mainstream process for atmospheric CO₂ removal and assist in combating global warming on a megaton scale.

Experimental section

Materials

ASTM D-1141-98 (13) Formula A (Lot: TL-ASTM-24061) Sea Salt was acquired from Lake Products Company LLC. Pure gas cylinders of CO₂ were obtained from Airgas®. Calcium oxide (CaO), magnesium oxide (MgO) (96%), manganese(II) oxide (MnO) (99%), titanium(II) oxide (TiO₂) (99.5% trace metals basis), iron(III) oxide (Fe₂O₃) (99.95% trace metals basis), aluminum oxide (Al₂O₃) (99% trace metals basis), and phosphorus(V) oxide (P₂O₅) (99.99% trace metals basis) were acquired from BeanTown Chemical. Hydrochloric acid (HCl) (A114C-212 Certified ACS Plus) was acquired from Fisher Scientific. Fisherbrand™ RT Basic Magnetic Stirrers were acquired from Fisher Scientific.

Simulated pea ridge mining waste

Simulated mining waste in solid form replicating tailings from the Pea Ridge Mine was made using the composition shown in Table 1. The

Table 1
Physical composition of simulated mining waste.

Component	REO (mg/L)	wt%
CaO	120,000	98
MgO	1500	1.2
MnO	300	0.3
TiO ₂	<100	0.1
Fe ₂ O ₃	<100	0.1
Al ₂ O ₃	<100	0.1
P ₂ O ₅	<100	0.1

recovered tailings are the result of rare earth extractions occurring at the Pea Ridge Mine targeting metals such as neodymium, dysprosium, lanthanum, and samarium. A recent publication from this group investigated recovering rare earth elements from mining tailings at the Pea Ridge Mine (Kang et al., 2026).

Real seawater collection

Seawater was collected from Sunset Beach in Treasure Island, FL in April 2025 at a ~1' sampling depth near the shoreline. The sample was filtered through a 2.7 µm Whatman 542 filter paper and stored at 3 °C for 60 days prior to any experimentation.

pH measurements

pH was measured using a VWR® pHenomenal® pH 1100 L meter. Prior to experimentation, the pH electrode was calibrated with buffer solutions representing pH 4, 7, and 10.

Alkaline additions

Simulated Pea Ridge mining waste was added to 1 L of ASTM seawater in varying concentrations to perform OAE. During alkaline addition, the pH meter was set to log measurements every 10 sec for 810 min.

Carbon dioxide gas purging system

Alkaline enhanced samples were met with a custom-assembled gas sparger to inject CO₂ gas at a flow rate of 3 NL·h⁻¹ and bring sample pH back to 8.2. During CO₂ sequestration, the pH meter was set to log measurements every 1 sec for 20 min.

Total inorganic carbon (TIC) analysis

TIC analysis was done with a CM5330 Acidification Module and CM5017 CO₂ Coulometer acquired from UIC, Inc. Samples of 10 mL were loaded into the cell at 25 °C and met with 10 mL of 2 M HCl. Ambient air was used for the carrier gas flowing at 100 mL·min⁻¹.

Total alkalinity (TA)

Total alkalinity measurements were performed using a TitraLab® AT1000 series workstation acquired from HACH®. Prior to measurement, the pH electrode was calibrated with buffer solutions representing pH 4, 7, and 10. Additionally, the titrant was calibrated with 0.005 M NaOH. 50 mL of each sample was mixed with 50 mL deionized water for several minutes prior to measurement.

X-ray diffraction (XRD)

Precipitated solids were filtered, dried at 20 °C, and characterized using a PANalytical Empyrean diffractometer with Cu Kα radiation (λ = 1.5406 Å). Patterns were collected over 5–90° 2θ with a step size of 0.026° and total scan time of 26 min. Phase identification was performed by comparison with reference patterns from the Materials Project database (aragonite mp-4626, calcite mp-3953, halite mp-22,862, brucite mp-30,247).

Gas absorption rate in seawater

Gaseous absorption rates for simulated mining waste in ASTM seawater were performed with a LI-850 CO₂/H₂O Gas Analyzer acquired from LI-COR®. 200 mL seawater samples were loaded into a 1 L glass jar attached to the gas analyzer by 43.5" of 1/8" ID perfluoro alkoxy plastic tubing. Total volume of the closed system (jar and tubing) is ~1.27 L. The gas analyzer logged CO₂ measurements every 1 min for 3 h.

Inductively coupled plasma optical emission spectroscopy

Elemental concentrations were analyzed by inductively coupled plasma optical emission spectroscopy (ICP-OES) using an iCAP Pro ICP-OES instrument (Thermo Scientific, Bremen, Germany). Prior to analysis, samples were gravimetrically diluted by factors of 10 to 1000 using 2 wt.% HNO₃ prepared from optima-grade nitric acid and ultrapure

water to ensure that analyte concentrations fell within the linear dynamic range of the instrument. The diluted solutions were introduced into the ICP-OES via an SC-2DXi autosampler (ESI, Omaha, NE, USA). The plasma was operated at 1150 W with an argon flow rate of 12 L min⁻¹, and all measurements were performed under axial viewing mode to maximize analytical sensitivity. Spectral acquisition and data processing were carried out using Qtegra™ Intelligent Scientific Data Solution™ software (Thermo Scientific). Elemental quantification was performed through external calibration using multi-element standards with concentrations ranging from 0.1 to 5 µg g⁻¹.

Temperature-controlled additions

Experiments in temperature-controlled environments were performed using a Professional Hotplate-Stirrer (Model 985VW0CHSUSP) acquired from VWR®. System temperature (5 °C, 10 °C) was controlled by a LabTech Recirculating Water Chiller (Model H50-500/115 V) acquired from LabTech, Inc. Water was continuously circulated through copper tubing resting in a Pyrex dish containing water surrounding a 1 L sample jar.

Results and discussion

Alkaline concentration study

This investigation aimed to determine the maximum solubility of simulated mining waste in 1 L of ASTM seawater to achieve the highest CO₂ sequestration without causing secondary precipitation. The pH stability, TIC, TA, and XRD measurements for these experiments are shown in Fig. 1. Fig. 1a highlights the time-dependent pH curves for simulated mining waste concentrations of 25, 50, 75, 100 and 125 mg·L⁻¹. Stable pH across long time scales generally indicates alkaline material has fully dissolved without precipitation. Unstable, decreasing pH curves across time could reveal secondary precipitation is occurring in the system. It can be seen from Fig. 1a that above concentrations of 75

mg·L⁻¹ (blue line), the pH begins to lose stability at time scales greater than one hour, indicating secondary precipitation. On the other hand, stability is observed in lower concentrations (25, 50, 75 mg·L⁻¹), indicating these being successful alkaline additions. Another major takeaway from these experiments is the correlation between maximum pH attained and concentration of alkaline material. The maximum pH and approximate equilibrium time for all measurements is shown in Table 2. Of mention, the pH values listed for concentrations 100 and 125 mg·L⁻¹ represent the highest pH reached before precipitation occurred. Upon precipitation, the pH for both solutions decreased for the remainder of the experiment and did not reach an equilibrium value.

Table 2

Summary table of all PRM alkaline additions in ASTM seawater. Y/N indicates whether the system experienced secondary precipitation. Values with ‘*’ represent the maximum pH before precipitation occurred and pH decreased. Values with ‘-’ represent solutions that precipitated, did not reach an equilibrium pH, and were not measured.

Concentration (mg·L ⁻¹)	25	50	75	100	125
pH Maximum	8.81	9.24	9.55	10.0*	10.1*
pH Post Absorption (equilibrium)	8.22	8.15	8.33	8.01	-
Equilibrium Time (min)	23	34	45	-	-
Precipitated	N	N	N	Y	Y
CO ₂ Sequestered (mg·L ⁻¹)	49.3 ± 0.7	76.7 ± 1	86.9 ± 1	-7.3 ± 2	-45.9 ± 3
ΔTotal Alkalinity Post Addition (mg CaCO ₃ ·L ⁻¹)	30.2 ± 5	64.2 ± 5	97.2 ± 6	-3.8 ± 5	-
ΔTotal Alkalinity Post Absorption (mg CaCO ₃ ·L ⁻¹)	41.7 ± 6	81.4 ± 5	108 ± 5	-	-

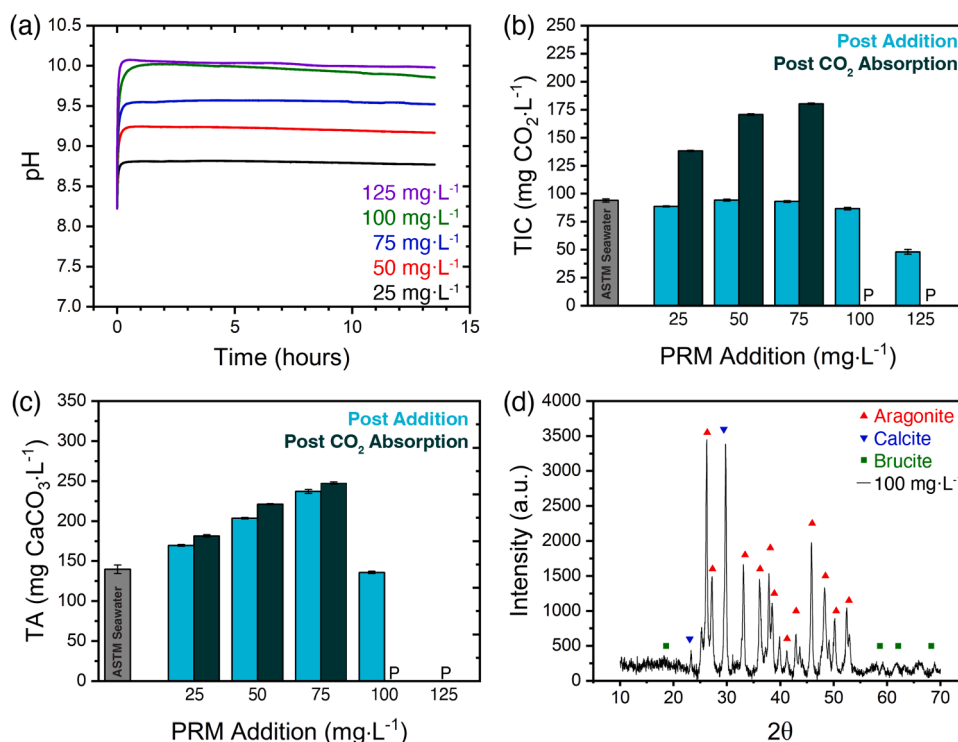


Fig. 1. Pea Ridge Mine alkaline waste additions in ASTM seawater. (a) pH of solution versus time for all concentrations at ambient temperature (20 °C) for 13.5 h. (b) TIC analysis for all concentrations. (c) TA analysis for all concentrations. (d) XRD of precipitate filtered out of solution at 100 mg·L⁻¹ concentration. Post addition measurements are represented in light blue bars. Post CO₂ absorption measurements are represented in dark blue bars. Concentrations with ‘P’ labels indicate the solution experienced secondary precipitation and were not measured. All error bars represent the standard deviation of (at least three) repeat measurements.

Upon addition of alkaline waste and demonstrated pH stability, the solutions were faced with a gas sparger to inject pure CO₂ into the solutions. A diagram of this gas absorption apparatus is shown in Fig. S1 where a custom gas sparger was made to fit into the 1 L jar and inject CO₂ into the enhanced samples for capture and storage. The goal of these experiments was to bring solution pH back to 8.2 and verify increases in CO₂ concentration that correlate with concentration of mining waste added. Note, in a real-world situation, the pH would be 0.1 higher than current ocean levels (pH≈8.1) and help reverse ocean acidification. The pH curves across time for all CO₂ absorptions performed are shown in Fig. S2 where the pH returned to 8.2 on the scale of a few minutes for all mining waste concentrations. Note, the absorption rates in these measurements were not analyzed as the valve for the gas sparger system is purely mechanical. A more controlled, automated system is needed to effectively evaluate these absorption rates and is warranted for further study.

Changes in inorganic carbon content for all alkaline additions were measured via TIC analysis as shown in Fig. 1b and Table 2. Measurements after alkaline addition are shown in light blue bars, while measurements after CO₂ absorption are shown in dark blue bars. The TIC of seawater varies from 85 to 104 ppm depending on the source and region. (Digdaya et al., 2020; Dickson, 2010; Prajapati and Singh, 2022) The ASTM seawater for this study was 94.0 ± 1.5 mg CO₂·L⁻¹ and is represented by a grey bar. To verify successful alkaline additions free of secondary precipitation, the TIC value after addition should remain the same as the starting solution (94.0 ± 1.5 mg CO₂·L⁻¹). When measuring TIC for this study, all solutions were left to rest without stirring for several hours to allow precipitated carbonates, if any, to settle towards the bottom of the jar. Stirring the solution agitates sediment (potentially carbonates) resting at the bottom which could potentially provide misleading TIC measurements. As seen in Fig. 1b, at concentrations 25, 50, and 75 mg·L⁻¹, the TIC value after addition remained unchanged from ASTM, potentially confirming successful alkaline additions. As the concentration was increased to 100 mg·L⁻¹, there was a noticeable decrease in TIC (-7.3 ± 1 mg CO₂·L⁻¹) from its starting value. Further increasing the concentration to 125 mg·L⁻¹ presented an even steeper decrease (-45.9 ± 3 mg CO₂·L⁻¹). These stark reductions in TIC, instabilities in pH, and visual solids formation in solution are indicative of secondary precipitation occurring above 75 mg·L⁻¹.

After bringing the solution pH back down to ~8.2 via CO₂ injection, these samples were evaluated for TIC increases corresponding to CO₂ absorption via HCO₃⁻ formation. Upon reaching equilibrium with ambient conditions, all solutions demonstrated stable pH close to 8.18 ± 0.13 as shown in Table 2. This return to 8.2 pH across all experiments indicates no additional CO₂ absorbing into solution upon consumption of all Ca(OH)₂ molecules. Based on stoichiometry, every mole of CaO can sequester two moles of CO₂ as previously shown in Eqs. (1) and 2. The highest TIC increase was 86.9 ± 0.7 mg CO₂·L⁻¹ at 75 mg·L⁻¹ concentration, slightly less than the theoretical maximum of 118 mg CO₂. Concentrations above 75 mg CO₂·L⁻¹ were not evaluated for TIC as precipitation had occurred. As such, these concentrations are labeled with a 'P' in Fig. 1b and c.

Changes in alkalinity for all concentrations are shown in Fig. 1c and Table 2. The TA of ASTM seawater for this study was 139.8 ± 5.3 mg CaCO₃·L⁻¹ and is represented by a grey bar. For this analysis, TA of the solution should increase upon addition of alkaline waste. As was the case with TIC measurements, all solutions were left to rest without stirring for several hours to allow any precipitated carbonates to settle towards the bottom of the jar and avoid misleading TA measurements. Increases in TA were observed (post addition) at concentrations 25, 50, and 75 mg·L⁻¹. The increases between concentrations were nearly linear, with each 25 mg of mining waste corresponding with ~30 mg increase in TA as CaCO₃. The largest post addition increase was observed at 75 mg·L⁻¹ where the TA increased by 97.2 ± 5 mg CaCO₃·L⁻¹. Once the concentration was increased to 100 mg·L⁻¹, the TA post addition showed a slight decrease. It is likely the sediment resting at the bottom of the jar was

precipitated CaCO₃. As such, TA measurements above 75 mg·L⁻¹ were not performed. After CO₂ absorption, these samples were evaluated for TA increases corresponding to CO₂ absorption via HCO₃⁻ formation. All concentrations below 100 mg·L⁻¹ showed increases in TA for post CO₂ absorption

Precipitated solids at 100 mg·L⁻¹ concentration were filtered and characterized by XRD as shown in Fig. 1d. The spectrum confirmed the presence of aragonite, (CaCO₃), calcite (CaCO₃), and brucite (Mg(OH)₂) crystal structures that wholly confirms secondary precipitation at this concentration. This study established 75 mg·L⁻¹ as the solubility limit for simulated mining waste in ASTM seawater that can theoretically sequester up to 118 mg of CO₂ and will be instrumental in the technoeconomic analysis presented later in this paper.

Gas absorption rate

The absorption rate of ambient CO₂ in alkaline enhanced seawater was measured via gas analyzer as previously described in the materials section. A diagram of this gas-liquid absorption system can be seen in Fig. S3. For this study, 200 mL of ASTM and real seawater from Sunset Beach, FL were loaded into a 1 L jar connected to the gas analyzer. Note, the starting CO₂ concentration in the system was ~420 ppm. Simulated mining waste concentrations free of secondary precipitation (25, 50, 75 mg·L⁻¹) were used for this study. Each concentration of simulated mining waste was added to the jar and stirred for three hours. Note, the control represents stock seawater (ASTM or Sunset Beach, FL). The results of these time-resolved absorption experiments are shown in Fig. 2 where the CO₂ concentration decreases across time for all concentrations, including the ASTM control. No changes in CO₂ were measured for the real seawater control. Exponential decay curves were fitted to the raw data for each time-dependent concentration used in this study. The amount of CO₂ absorbed, along with maximum absorption rates in both ppm and mg of CO₂ per hour, are shown in Table 3. As demonstrated from the table, it is clear that alkaline mining waste had a direct correlation with the amount of absorbed CO₂ and rate of absorption. The lowest concentration (25 mg·L⁻¹) of mining waste showed similar absorption behavior in ASTM and real seawater. The control experiments absorbed minimal to no CO₂ in three hours, while the highest concentration (75 mg·L⁻¹) in ASTM absorbed 0.67 mg. TIC analysis of the seawater from the 75 mg·L⁻¹ experiment revealed a direct increase of 0.58 mg CO₂ at a rate of 0.46 mg CO₂ per hour. This value is close to what was measured by the gas analyzer and confirms absorption of CO₂. Note, this system contains approximately 0.83 mg of CO₂ before the start of each experiment.

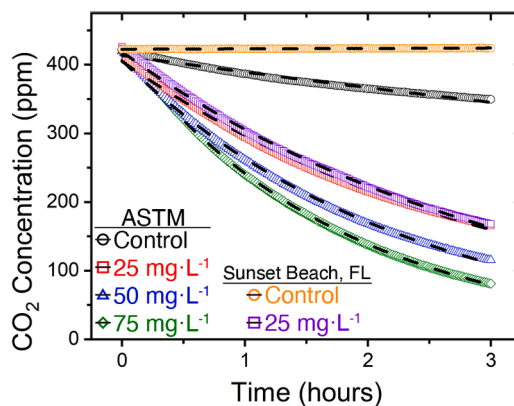


Fig. 2. Absorption of CO₂ for varying mining waste concentrations in ASTM and real seawater retrieved from Sunset Beach, FL. All measurements performed at ambient temperature (20 °C) for 3 h. R² value for all exponential fittings is greater than 0.99.

Table 3

Summary of CO₂ absorption rates for varying concentrations of alkaline mining waste in seawater.

Seawater	Control (0 mg·L ⁻¹)		25 mg·L ⁻¹		50 mg·L ⁻¹		75 mg·L ⁻¹	
	ASTM	Real	ASTM	Real	ASTM	Real	ASTM	Real
CO ₂ Absorbed (mg)	0.140	0	0.51	0.50	0.60	0.60	0.67	0.67
Max Absorption Rate ($\frac{\text{ppm}}{\text{hour}}$)	24.8	0	125	132	176	176	234	234
Max Absorption Rate ($\frac{\text{mg CO}_2}{\text{hour}}$)	0.05	0	0.25	0.26	0.34	0.34	0.46	0.46

Real seawater concentration study

Seawater was obtained from Sunset Beach in Treasure Island, FL to compare with ASTM seawater. A comparison of TIC and TA values for both stock seawaters is shown in Fig. S4. When comparing TIC values, ASTM ($94.0 \pm 1.5 \text{ mg CO}_2\text{L}^{-1}$) is $4.4 \pm 2.4 \text{ mg CO}_2\text{L}^{-1}$ greater than the real seawater ($89.6 \pm 1.9 \text{ mg CO}_2\text{L}^{-1}$). For TA values, ASTM ($139.8 \pm 5.3 \text{ mg CaCO}_3\text{L}^{-1}$) is $16.1 \pm 5.8 \text{ mg CO}_2\text{L}^{-1}$ greater than real seawater ($123.7 \pm 2.3 \text{ mg CaCO}_3\text{L}^{-1}$). Another metric of mention is their difference in pH, with real seawater (pH $\approx$ 8.1) being slightly more acidic than ASTM (pH=8.2). ICP-OES was performed to compare the compositions of both solutions. These compositions are displayed in Tables S1 and S2. When observing these tables, it is noticeable that their compositions are very similar. With these differences noted, a concentration study was

performed with simulated mining waste in real seawater to determine the solubility limit and maximum CO₂ storage as shown in Fig. 3 and Table 4. An overlay of pH stability curves for various simulated mining waste concentrations (25, 50, 75 mg·L⁻¹) in Sunset Beach (light to dark blue curves) and ASTM (light to dark red curves) seawater are shown in Fig. 3a. The pH curves for real seawater are shown standalone in Fig. S5.

Table 4

Summary table of all PRM alkaline additions in real seawater from Sunset Beach, FL. Y/N indicates whether the system experienced secondary precipitation. Values with ‘*’ represent the maximum pH before precipitation occurred and pH decreased. Values with ‘-’ represent solutions that precipitated, did not reach an equilibrium pH, and were not measured.

Concentration (mg·L ⁻¹)	25	50	75
pH Maximum	8.90	9.40*	10.0*
pH Post Absorption (equilibrium)	8.37	–	–
Equilibrium Time (min)	63	–	–
Precipitated	N	Y	Y
CO ₂ Sequestered (mg·L ⁻¹)	33.0 $\pm$ 2	–42.2 $\pm$ 2	–71.8 $\pm$ 2
Δ Total Alkalinity Post Addition (mg CaCO ₃ ·L ⁻¹)	43.5 $\pm$ 5	–62.5 $\pm$ 2	–81.2 $\pm$ 2
Δ Total Alkalinity Post Absorption (mg CaCO ₃ ·L ⁻¹)	38.5 $\pm$ 6	–	–

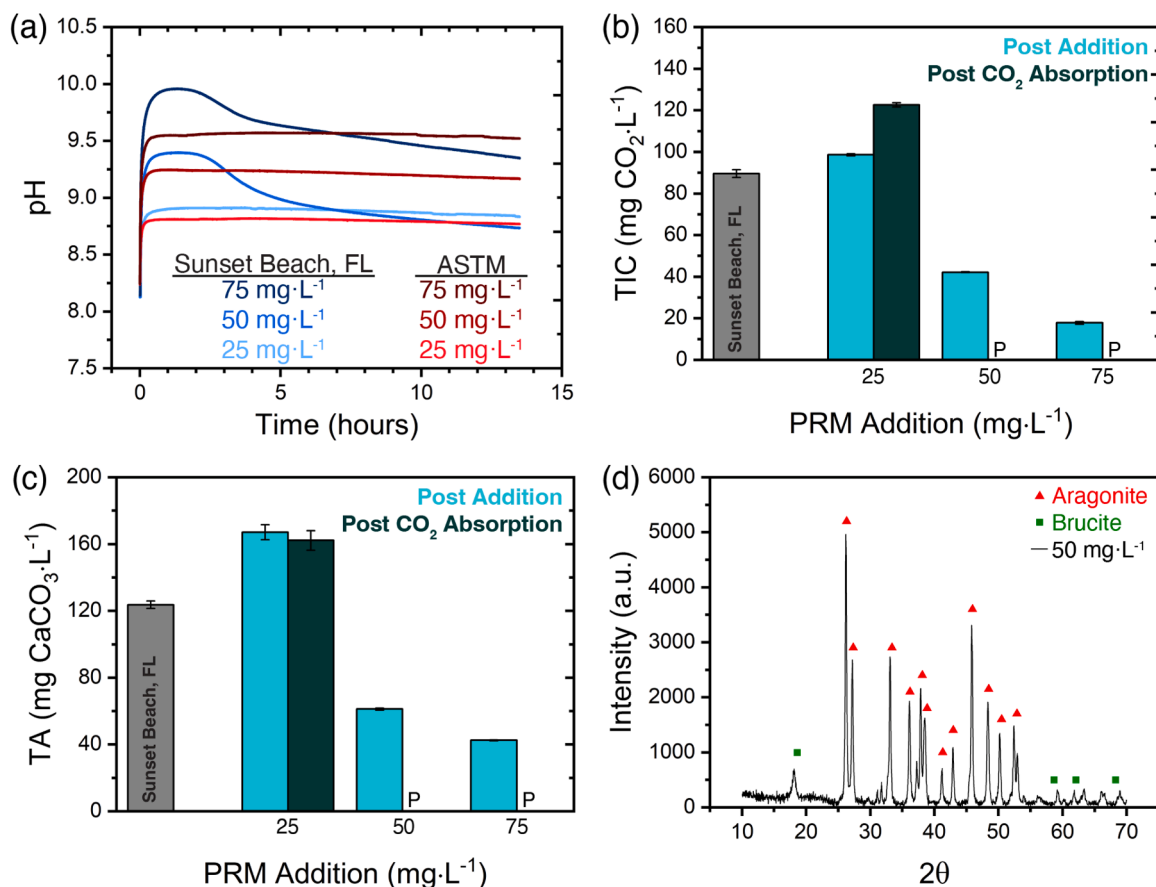


Fig. 3. Pea Ridge Mine alkaline waste additions in real seawater from Sunset Beach, FL. (a) pH of solution versus time for all concentrations at ambient temperature (20 °C) for 13.5 h. (b) TIC analysis for all concentrations. (c) TA analysis for all concentrations. (d) XRD of precipitate filtered out of solution at 50 mg·L⁻¹ concentration. Post addition measurements are represented in light blue bars. Post CO₂ absorption measurements are represented in dark blue bars. Concentrations with ‘P’ labels indicate the system experienced secondary precipitation and were not measured. All error bars represent the standard deviation of (at least three) repeat measurements.

Real seawater behaved much differently than ASTM, with real seawater only demonstrating stable pH behavior across 13.5 h at 25 mg·L⁻¹. The stark downturns in pH seen in 50 and 75 mg·L⁻¹ signify secondary precipitation occurring. When considering the solubility limits referenced in other investigations with real seawater, (Kowalczyk et al., 2024; Moras et al., 2022) this concentration (25 mg·L⁻¹) aligns well. It is ostensible that ASTM seawater is not truly representative of seawater from Sunset Beach. While it is unclear at the time of this study, we believe the differences in solubility are dependent on discrepancies in TIC, TA, pH, dissolved organic matter, and biological matter. The maximum pH and equilibrium time for all real seawater measurements is shown in Table 4. Note, the max pH values for concentrations of 50 and 75 mg·L⁻¹ represent the highest pH reached before precipitation occurred.

TIC analysis was performed to evaluate changes in inorganic carbon content as shown in Fig. 3b and Table 4, where measurements were taken after alkaline addition (light blue bars) and after CO₂ absorption (dark blue bars). The real seawater for this study (89.6 ± 1.9 mg CO₂·L⁻¹) is represented by a grey bar. It can be seen in Fig. 3b that after addition of mining waste, only the lowest concentration (25 mg·L⁻¹) held a (relatively) constant TIC value. The slight increase in TIC could be a result of the alkaline waste sequestering some ambient CO₂ immediately upon dissolution. As the concentration was increased above 25 mg·L⁻¹, steep TIC decreases were measured that indicate secondary precipitation.

Upon addition of alkaline waste and demonstrated pH stability, the solutions were injected with pure CO₂ as previously described. The pH curve across time for CO₂ absorption at 25 mg·L⁻¹ concentration is shown in Fig. S6 where the pH returned to 8.2 on the scale of minutes. Upon reaching equilibrium with ambient conditions, the solution demonstrated stable pH close to 8.37 as shown in Table 4. At this concentration, there were ~33 mg of CO₂ sequestered. Concentrations above 25 mg CO₂·L⁻¹ precipitated and are represented by a 'P' in Fig. 3b and c.

Changes in alkalinity for all concentrations are shown in Fig. 3c and Table 4. The TA of real seawater (123.7 ± 2.3 mg CaCO₃·L⁻¹) is represented by a grey bar. Increases in alkalinity were only observed (post addition) at the lowest concentration (25 mg·L⁻¹), where TA increased by 43.4 ± 5.0 mg CaCO₃·L⁻¹. Once the concentration was increased to 50 and 75 mg·L⁻¹, the TA (post addition) showed a significant decrease. It is apparent the newly formed sediment resting at the bottom of the jar was precipitated CaCO₃. After CO₂ injection, the lowest concentration (25 mg·L⁻¹) showed no significant change in TA, unlike the increases seen in the ASTM study. This difference in TA trends before and after CO₂ absorption for ASTM and real seawater are unclear and warranted for further investigation.

Precipitated solids at 50 mg·L⁻¹ concentration were filtered and characterized by XRD as shown in Fig. 3d where the presence of aragonite and brucite crystal structures were confirmed. This study established 25 mg·L⁻¹ as the solubility limit for simulated mining waste in real seawater from Sunset Beach, FL that can theoretically sequester up to 39 mg of CO₂.

Temperature impact on solubility

Many investigations in the literature have established temperature playing a role on the solubility of CaO in water. (Moras et al., 2022; Moras et al., 2024) The dissolution of CaO exhibits retrograde solubility, where solubility increases with decreasing temperature. This means if the proposed process were carried out in colder climates (i.e., Hudson Bay (Galbraith and Larouche, 2011)), it is possible to dissolve higher concentrations of CaO and capture more CO₂ on a volume basis. (Zeebe and Wolf-Gladrow, 2001) We must take into consideration that the Hudson Bay has increased by 1 °C in the last 30 years (Stroeve et al., 2024) which could impact the effectiveness of OAE in colder climates as global warming continues. It was previously established that 75 mg·L⁻¹

is the solubility limit (free of secondary precipitation) for simulated mining waste in ASTM seawater at ambient temperature (20 °C). A concentration study was performed with simulated mining waste in ASTM seawater at colder temperatures (5, 10 °C). A diagram of the system used for these temperature-controlled studies is shown in Fig. S7. The full analysis for 10 °C ASTM seawater is shown in Fig. 4 and Table 5. The TIC (94.0 ± 1.5 mg CO₂·L⁻¹) and TA (139.8 ± 5.3 mg CaCO₃·L⁻¹) of ASTM seawater are represented by a grey bar in Fig. 4b and c. Fig. 4a presents the pH stability curves across 13.5 h for mining waste concentrations: 100, 125, 150 and 200 mg·L⁻¹. At first glance, all concentrations appear to have stable pH behavior across the entirety of the experiment. However, a closer examination of the first few hours (image subset) reveals the highest concentration (200 mg·L⁻¹, purple line) appearing to decrease steeper than other concentrations at the same time. The maximum pH and equilibrium time for all 10 °C measurements is shown in Table 5. Note, the max pH value at 200 mg·L⁻¹ represents the highest pH before precipitation occurred.

Changes in inorganic carbon content were again measured via TIC analysis as shown in Fig. 4b and Table 5, where measurements were taken after alkaline addition (light blue bars) and CO₂ absorption via gas sparger (dark blue bars). It is observed from Fig. 4b that concentrations 100, 125, and 150 mg·L⁻¹ held (relatively) constant TIC values unchanged from ASTM seawater. When the addition was increased to 200 mg·L⁻¹, the post addition TIC drastically decreased.

These 10 °C alkaline enhanced solutions were then injected with pure CO₂ as seen in the other studies. The CO₂ absorption pH curves across time for all concentrations are shown in Fig. S8. Upon reaching equilibrium with ambient conditions, all solutions demonstrated stable pH close to 8.32±0.07 as shown in Table 5. The most CO₂ sequestered was 165±3 mg CO₂·L⁻¹ at a concentration of 150 mg·L⁻¹. Above concentration 150 mg CO₂·L⁻¹, much sediment appeared to precipitate in solution and were not evaluated for TIC. As such, this concentration has a 'P' above in Fig. 4b and c.

Changes in alkalinity at 10 °C are shown in Fig. 4c. Increases in alkalinity were observed for post addition and absorptions at concentrations of 100, 125, and 150 mg·L⁻¹. These observed alkalinity increases are summarized in Table 5.

The precipitation occurring at a concentration 200 mg·L⁻¹ was filtered and characterized by XRD as shown in Fig. 4d. The presence of aragonite, calcite, and brucite confirms secondary precipitation at this concentration. When recalling the slight TIC decrease measured for 150 mg·L⁻¹ concentration, it is possible precipitation is occurring at this concentration. However, the precipitation generated at this concentration was not enough to measure XRD. Further investigation is warranted for this concentration at larger scales (>1 L) to generate more precipitation and complete XRD analysis. This study established 150 mg·L⁻¹ as the solubility limit for simulated mining waste in ASTM seawater at 10 °C that can potentially sequester 314 mg of CO₂.

System temperature was then lowered to 5 °C for another ASTM concentration study shown in Fig. 5 and Table 6. Fig. 5a presents the pH stability curves for simulated mining waste concentrations: 150, 200, and 250 mg·L⁻¹. When taking a closer look at the first few hours (image subset) of the experiment, it appears the pH of the highest concentration (250 mg·L⁻¹, green line) decreases steeper than other concentrations. The maximum pH and equilibrium time for all 5 °C measurements is shown in Table 6.

Changes in inorganic carbon content were again characterized by TIC analysis as shown in Fig. 5b, where measurements were taken after alkaline addition (light blue bars) and CO₂ absorption via gas sparger (dark blue bars). The TIC values for post addition at 150 and 200 mg·L⁻¹ held consistent values with ASTM seawater. Increasing the concentration to 250 mg·L⁻¹ lead to a drastic decrease (-44 mg CO₂·L⁻¹) in TIC. The amount of CO₂ sequestered for each addition at 5 °C is shown in Table 6.

Alkaline enhanced solutions at 5 °C were then injected with pure CO₂ as previously described. The pH curves across time for all CO₂

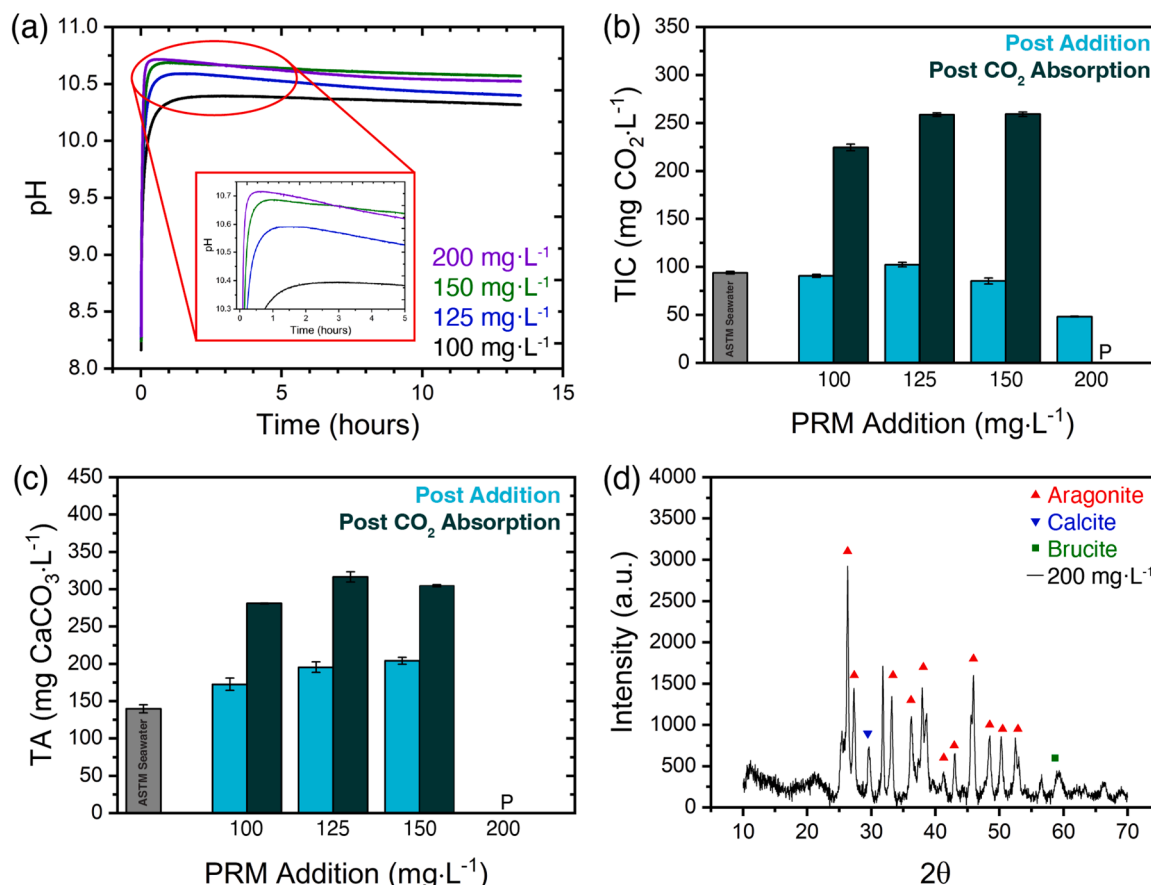


Fig. 4. Pea Ridge Mine alkaline waste additions in ASTM seawater at 10 °C. (a) pH of solution versus time for all concentrations at 10 °C for 13.5 h. (b) TIC analysis for all concentrations. (c) TA analysis for all concentrations. (d) XRD of precipitate filtered out of solution at 200 mg·L⁻¹ concentration. Post addition measurements are represented in light blue bars. Post CO₂ absorption measurements are represented in dark blue bars. Concentrations with 'P' labels indicate the system experienced secondary precipitation and were not measured. All error bars represent the standard deviation of (at least three) repeat measurements.

Table 5

Summary table of all PRM alkaline additions in ASTM seawater at 10 °C. Y/N indicates whether the system experienced secondary precipitation. Values with '*' represent the maximum pH before precipitation occurred and pH decreased. Values with '-' represent solutions that precipitated, did not reach an equilibrium pH, and were not measured.

Concentration (mg·L ⁻¹)	100	125	150	200
pH Maximum	10.4	10.6	10.7	10.8*
pH Post Absorption (equilibrium)	8.39	8.26	8.32	–
Equilibrium Time (min)	125	175	175	–
Precipitated	N	N	N	Y
CO ₂ Sequestered (mg·L ⁻¹)	131±4	165±2	165±3	–45.7 ± 2
ΔTotal Alkalinity Post Addition (mg CaCO ₃ ·L ⁻¹)	32.7 ± 10	55.8 ± 9	64.4 ± 7	–
ΔTotal Alkalinity Post Absorption (mg CaCO ₃ ·L ⁻¹)	141±5	177±5	165±5	–

absorptions are shown in Fig. S9. Upon reaching equilibrium with ambient conditions, all solutions demonstrated stable pH in range of 8.3 to 8.4 as shown in Table 6. The highest CO₂ sequestered was 221±3 mg CO₂·L⁻¹ at a concentration of 200 mg·L⁻¹. Concentrations above 200 mg CO₂·L⁻¹ precipitated and have a 'P' label in Fig. 5b and c. The measured increases in alkalinity for post addition and absorption are also shown in

Table 6.

The precipitated solids at a concentration 250 mg·L⁻¹ were characterized by XRD as shown in Fig. 5d, where the presence of aragonite, calcite, and brucite were confirmed and indicative of secondary precipitation. This study established 200 mg·L⁻¹ as the solubility limit for simulated mining waste in ASTM seawater at 5 °C that can potentially sequester 392 mg of CO₂.

Performing temperature studies with real seawater from Sunset Beach, FL will confirm the effect temperature has on solubility. Further work is warranted to verify the changes observed with ASTM seawater.

Techno-economic analysis (TEA)

To evaluate the economic feasibility of this proposed process, an industrial-scale TEA was conducted for the dissolution of simulated mining waste into ocean waters and subsequent CO₂ sequestration. The solubility limits established earlier for ASTM (75 mg·L⁻¹, ~1 hour dissolution) and real seawater (25 mg·L⁻¹, ~1 hour dissolution) were used as the basis for this process scale-up. Upon dissolution, 1 L of ASTM seawater can sequester up to 118 mg of CO₂ (39 mg CO₂ for real seawater) when following a 2:1 molar basis. The scale for this process is 50,000 short tons of simulated mining waste annually that will sequester ~71,200 t of CO₂. A general summary of the concentration and scale for this process is summarized in Table 7.

As previously mentioned, this model involves injecting pure CO₂ from another capture process (direct air capture, point source) for permanent storage in the discharged seawater. This utilization of previously captures CO₂ provides an alternative to compressing unusable gas into

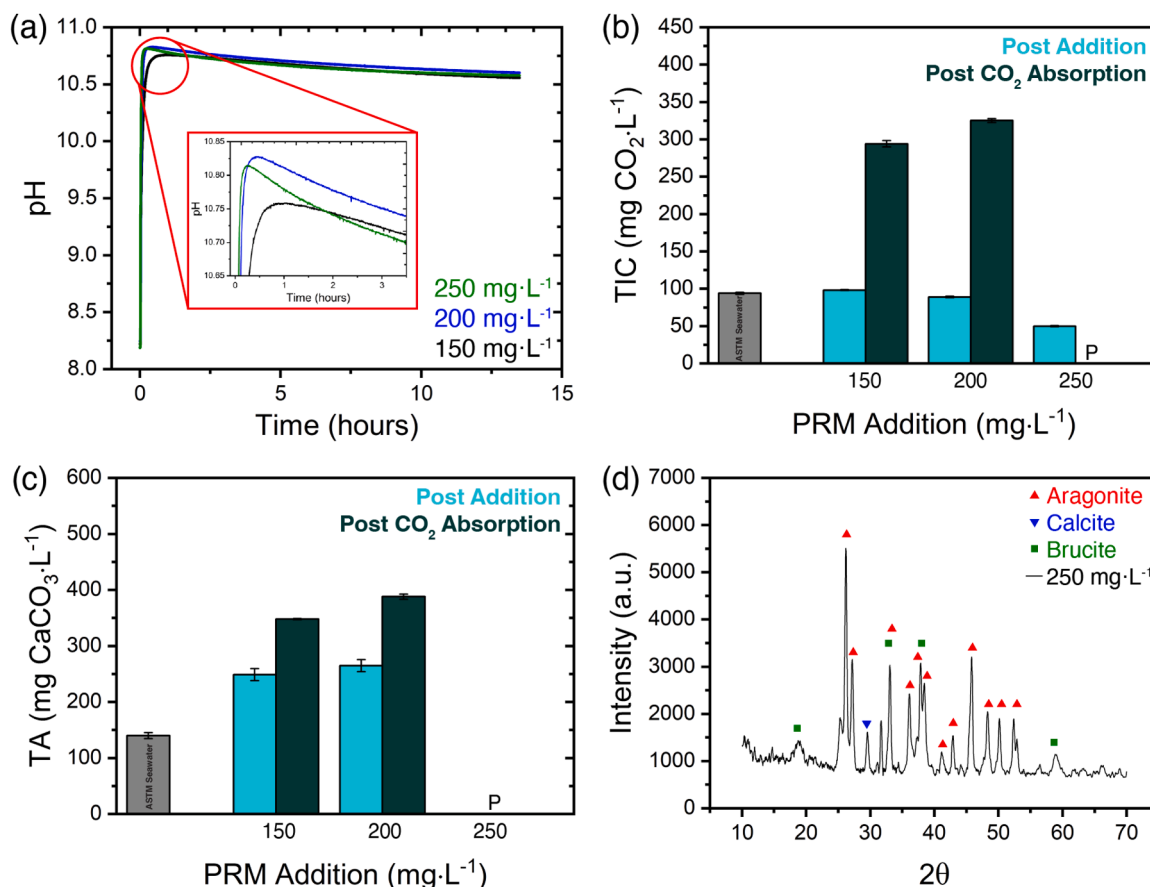


Fig. 5. Pea Ridge Mine alkaline waste additions in ASTM seawater at 5 °C. (a) pH of solution versus time for all concentrations at 5 °C for 13.5 h. (b) TIC analysis for all concentrations. (c) TA analysis for all concentrations. (d) XRD of precipitate filtered out of solution at 250 mg·L⁻¹ concentration. Post addition measurements are represented in light blue bars. Post CO₂ absorption measurements are represented in dark blue bars. Concentrations with 'P' labels indicate the system experienced secondary precipitation and were not measured. All error bars represent the standard deviation of (at least three) repeat measurements.

Table 6

Summary table of all PRM alkaline additions in ASTM seawater at 5 °C. Y/N indicates whether the system experienced secondary precipitation. Values with '*' represent the maximum pH before precipitation occurred and pH decreased. Values with '-' represent solutions that precipitated, did not reach an equilibrium pH, and were not measured.

Concentration (mg·L ⁻¹)	150	200	250
pH Maximum	10.75	10.83	10.81*
pH Post Absorption (equilibrium)	8.30	8.42	–
Equilibrium Time (min)	~200	~200	–
Precipitated	N	N	Y
CO ₂ Sequestered (mg·L ⁻¹)	200±4	221±3	–44.0 ± 2
ΔTotal Alkalinity Post Addition (mg CaCO ₃ ·L ⁻¹)	109±10	125±9	–
ΔTotal Alkalinity Post Absorption (mg CaCO ₃ ·L ⁻¹)	208±5	248±7	–

an underground class VI well. As such, the energy requirements and processing costs presented here do not incorporate the external capture process and strictly focus on mixing simulated mining waste with seawater and injection of gas.

The proposed mixing facility tailored for ASTM seawater will include mixing tanks, agitators, pumps, grain elevators, conveyor belts, hoppers

Table 7

Overview of basis for mixing plant and annual processing values.

Parameter	Value	Unit
Solubility	75	mg·L ⁻¹
Equilibrium Time	1	Hour
pH Maximum	9.6	pH
Mining Waste Processed Annually	50,000	Short tons
Water Processed Annually	160	Billion gallons
CO ₂ Sequestered Annually	71,200	Metric tons

and silos. For simplicity, a predetermined amount of equipment will occupy what will be referred to as one "unit" of the facility. In total, 38 units will continuously operate in the facility. A summary of all equipment in each unit, along with the entire process are shown in Table 8. A process flow diagram showing all equipment needed to operate one-half of a unit is shown in Fig. S10.

Each unit will contain 12 100,000-gallon galvanized steel mixing

Table 8

Equipment totals in proposed mixing facility.

Equipment	Total (per unit)	Total (entire process)
Mixing Tanks	12	456
Agitators (2 per tank)	24	912
Pumps	2	76
Grain Elevators	1	38
Conveyor Belts	3	114
Hoppers	2	76
Silos		3

tanks operating at 80% capacity. The process will begin by pumping seawater into the tanks and where it is met with mining waste discharged via hopper. The hoppers are fed by conveyor belts connected to a grain elevator that elevates the solids waste above the mixing tanks (23 ft). The mining waste mixes with seawater for ~1 hour to ensure full dissolution, supported from our lab-scale measurements. The highly alkaline seawater is then injected with CO₂ to lower solution pH back to 8.2. The seawater is then drained from the bottom of the tank and released back into the ocean via piping. The residence time for this batch process is 2 h (filling time, mixing, gas absorption, and draining of seawater). This means that at any given time, the plant is actively processing 36 million gallons of seawater with ~10 tons of solid mining waste.

Each mixing tank is equipped with two 50,000-gallon capacity agitators to mix mining waste with seawater. Specifications of importance for mixing tanks and agitators are shown in Table S3. The agitators rotated at 70% of their maximum velocity to follow industrial mixing practices. (ProQuip 2018) We recognize there are larger agitators capable of mixing 80,000 gallons of water, but chose to focus on incorporating equipment into this process that had accessible prices online, without needing a quote. To determine an energy requirement for the mixing process, the following shaft power equation (Tools) was used:

$$P = N_p \rho N^3 D^5$$

Using the variables described in Table 9, each agitator requires 1.5 kW (2 hp) of power.

For each unit, 1 p.m. will be connected to the entrapment for filling the mixing tanks (12) to 80% capacity. The second pump is mounted to a pipe pulling from the bottom of the mixing tanks for draining purposes. Information pertaining to pumps and piping is shown in Table S4.

Solids processing equipment for this facility includes silos, grain elevators, conveyor belts, and hoppers. The yearly supply of solid mining waste (50,000 short tons) will be stored across three 150,000-bushel sized silos. The solid waste will be moved from silos to mixing tanks via grain elevators, conveyor belts, and hoppers. The specifications for all solids processing equipment are shown in Table S5.

An energy requirement for this process was then calculated taking into account all electricity-dependent equipment (grain elevators, conveyor belts, agitators, and pumps). A breakdown of the energy consumption by equipment type is shown in Table S6. The total energy requirement for this proposed (ASTM seawater) process is 3257 kW, or 1.4 GJ per ton of CO₂ stored annually. If this facility were to process real seawater, this energy requirement would increase to 4.2 GJ per ton of CO₂.

Sizing of the facility, installation estimates, and hourly rates for operators are shown in Table S7. The size of the warehouse was estimated to be 26.8 acres, with installation taking approximately 2.6 years. The plant life for this facility is 40 years. (Renforth and Kruger, 2013)

Once all assumptions were taken into consideration, the capital (CapEx) and operating (OpEx) expenditures for this ASTM seawater-based process were compiled. The cost of electricity used for this analysis is \$0.1275 per kWh. An overview of the CapEx (1/10th of total cost) and OpEx are shown in Table 10 and Fig. 6, where a total processing cost was estimated to be \$266.34 per ton of CO₂ sequestered. For a real seawater facility, this processing cost would raise to \$807 per ton of CO₂

Table 9
Variables needed for agitator power requirement calculation.

Quantity	Variable	Value	Unit
Shaft Power Required	P		Watts
Power Number	N _p	0.3	
Impeller Diameter	D	1.5	meters
Rotational Speed	N	0.8	rev/s
Liquid Density	ρ	1000	kg/m ³

Table 10

Total capital and operating costs for proposed process.

Components	Cost
CapEx	
Agitators	\$36,122,000
Mixing Tanks	\$31,177,000
Warehouse Construction	\$23,305,000
Installation	\$6840,000
Pumps	\$4703,000
Piping & Fittings	\$2995,000
Conveyor Belts	\$1784,000
Silos	\$831,000
Grain Elevators	\$524,000
Hoppers	\$220,000
Total Capital Costs	\$108,501,000
OpEx	
Electricity	\$3641,000
Labor	\$3201,000
Maintenance	\$1266,000
Total Operating Costs	\$8108,000
TOTALS	
Gross Processing Cost	\$116,609,000
CapEx (\$/ton CO ₂)	\$152.43
OpEx (\$/ton CO ₂)	\$113.91
Processing Cost (\$/ton CO₂)	\$266.34

due to the difference in solubilities discovered in lab-scale measurements. The capital costs were put on a 10-year payback period. (Huang et al., 2025; Zhang et al., 2015) We recognize that the TEA is mostly centered around ASTM seawater which showed higher simulated mining waste solubility than real seawater. This staggering difference between ASTM and real seawater is warranted for further investigation, where perhaps other regions of the world contain seawater that demonstrates a mining waste solubility closer to that of ASTM. As mentioned previously, we believe the differences in solubility are dependent on discrepancies in TIC, TA, pH, total dissolved solids, dissolved organic matter, and biological matter.

The proposed TEA model was then used to calculate an energy requirement and processing cost for all studies performed in this investigation. A summary of these values is shown in Table 11.

The proposed process is quite large, and many things must be taken into consideration during design and implementation. Scale up of any process can lead to issues that were not present in lab-scale experimentation. Some of the issues that must be given attention include hydraulic loss in all pumps throughout the facility, and the presence of fouling in the piping and tanks. The residence time chosen for the tanks may also create issues when considering the mixing volume is increasing from 1 L to 100,000 gallons. Challenges might also arise in the solids processing of this facility when considering potential variability in particle size and agglomeration of waste. Finding coastal land to build this seawater intake and outfall facility and the constraints that come with working near a shoreline may also bring difficulties. Something to also consider is the different compositions of high purity CO₂ that will exist depending on source of capture (flue gas, air, etc.).

We recognize that this model is mirroring lab-scale results that used simulated mining waste, and not real-life tailings. Further research is warranted with real mining waste containing trace amounts of metals that may be harmful to the environment and create issues with fouling and equipment. The real-life tailings may also have different dissolution kinetics that could alter the residence time for mixing tanks. This discharge point will also create an intersection of fresh seawater and water leaving the facility with an elevated inorganic carbon content. While we anticipate precipitation will not occur, further research is necessary to examine this mixing at millions of gallons scale to analyze the potential for delayed precipitation. Incorporating advanced modeling into this design process could also help alleviate concerns with precipitation occurring.

To compare this proposed process with other investigations, a literature search was performed to compile solubilities and energy requirements from other OAE studies. These findings are summarized in Table 12. Note, when considering different compositions of seawater,

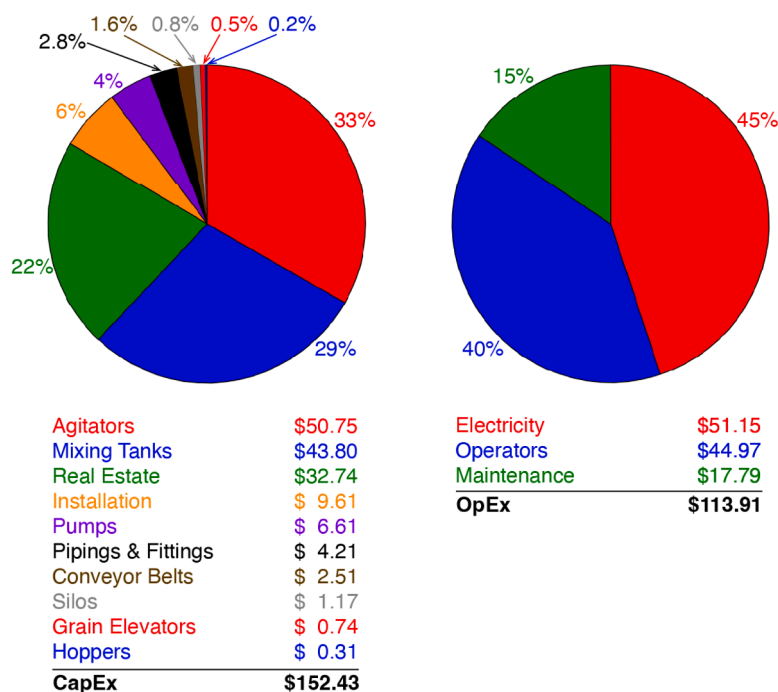


Fig. 6. Cost distribution of proposed mixing facility (in \$ per ton of stored CO₂) for dissolution of mining waste and sequestration of CO₂: (a) CapEx and (b) OpEx.

Table 11

Summary of energy requirement and processing cost for all studies in this paper.

Seawater	Temperature (°C)	Solubility	Energy Requirement (GJ/ton of CO ₂)	Processing Cost (\$/ton of CO ₂)
ASTM	25	75	1.4	266
ASTM	25	50	2.1	399
ASTM	10	125	0.89	162
ASTM	10	150	0.75	135
ASTM	5	150	0.75	135
ASTM	5	200	0.58	103
Sunset Beach, FL	25	25	4.2	807

alkaline feedstock, and varying system parameters, it can be challenging to truly compare each individual study.

Conclusion

This work investigates the use of simulated mining waste from traditional metal (e.g., critical materials) mining industries to perform OAE. Incorporating highly alkaline mining waste into OAE is favorable compared to common ocean liming processes that utilize the calcination of limestone. Concentration studies were performed for mining waste in ASTM and real seawater from Sunset Beach, FL to evaluate the solubility limit and maximum CO₂ that can be sequestered on a volume basis. Notably, this study brought to light the drastic differences between ASTM and real seawater with different solubility limits of 75 mg·L⁻¹ and 25 mg·L⁻¹, respectively. Temperature studies were performed to evaluate the effectiveness of OAE in colder climates where the solubility of mining waste doubled at 5 °C. One of the biggest limitations of OAE is the solubility limit of CaO, and further research is warranted to increase solubility and improve the capture performance of this process. Future directions will involve testing this process with real mining tailings and improving the dissolution rate of these tailings by shrinking particle size (s) to lessen residence times in the proposed mixing facility. Incorporating PHREEQ or CO2SYS modelling into this research will also

Table 12

Literature comparison for various OAE processes.

OAE Approach	Simulation/ Experimental	Solubility	Energy Requirement (GJ/ton of CO ₂)	Reference
Steel slag in seawater with flue gas	Simulation	N/A	1.02	(Li et al., 2020)
Calcined CaO via shipping freighters in ocean	Simulation	N/A	6.1	(Renforth et al., 2013)
Mineral carbonation and ocean liming with MgO	Simulation	N/A	7.1	(Renforth and Kruger, 2013)
Slaked lime in seawater from Livorno, Italy	Both	5–10 mg·L ⁻¹	N/A	(Varliero et al., 2024)
Ocean liming in seawater from Broken Head, New South Wales, Australia	Both	15–30 mg·L ⁻¹	N/A	(Moras et al., 2022)
Calcined CaO via shipping freighters in ocean	Simulation	20 mg·L ⁻¹	4.8–7.2	(Kowalczyk et al., 2024)
Alkaline mining waste via mixing facility (ASTM or Sunset Beach, FL)	Experimental	25–75 mg·L ⁻¹	1.4–4.2	This work

enhance the future of this research in avoiding precipitation. These tailings could also serve useful in carbon mineralization techniques that can potentially be used to capture CO₂ from air and flue gas.

This study proposes a new approach to OAE with a closed loop system of mixing and absorption that is more cognizant of changes in acidity on aquatic life and eliminates uncertainties that come with releasing hydroxides into open ocean waters. This proposed process has

an energy requirement of 1.4 GJ per ton of stored CO₂, along with a processing cost of ~\$266 per ton of CO₂. Identifying additional sources of alkaline waste can help OAE grow as a carbon capture strategy. Results from this study underscore the importance of identifying sustainable sources of alkaline materials to continue advancing OAE research and assist in combating global warming.

CRedit authorship contribution statement

Nicholas Gregorich: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Funding acquisition. **Joseph Kalla:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation. **Mary Hannah Irwin:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation. **Saurabh Prakash Pethe:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation. **Mustafa Abbas:** Methodology, Investigation, Formal analysis, Data curation. **Zachary Coin:** Methodology, Investigation, Formal analysis, Data curation. **Mariappan Parans Paranthaman:** Validation, Supervision. **Majbah Uddin:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis. **Shailesh Dangwal:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization. **Ramesh Bhawe:** Validation, Supervision. **Syed Z. Islam:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ccs.2026.100629](https://doi.org/10.1016/j.ccs.2026.100629).

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